



Minor Project Report

submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY
in
COMPUTER SCIENCE & TECHNOLOGY

by

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Under the guidance of

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Candidate's Declaration

I hereby certify that the project work entitled "**Adaptive Distance Weighted K-Nearest Neighbors for High-Dimensional Classification**" submitted in partial fulfillment of the requirements for the degree of B.Tech is an authentic record of my work carried out under the supervision of Touseef Iqbal.

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Acknowledgement

We would like to express our sincere gratitude to our project guide for their continuous support, valuable guidance, and encouragement throughout the development of this project.

We also thank the School of Computer Science, UPES, for providing the necessary resources and environment to carry out this project. We are grateful to our faculty members and peers for their suggestions and constructive feedback.

Finally, we would like to thank our parents and friends for their constant motivation and support.

Abstract

This project focuses on improving the performance of classification algorithms using an enhanced version of the K-Nearest Neighbors (KNN) technique. Traditional KNN models often suffer from limitations such as equal weighting of neighbors and sensitivity to the choice of K. To address these issues, this project implements and compares multiple variants including Baseline KNN, Distance Weighted KNN, and an Adaptive Distance Weighted KNN model.

The dataset used in this study is a binary classification dataset for predicting class labels based on multiple features. Various preprocessing steps such as feature selection, normalization, and correlation analysis were performed to improve model efficiency.

The performance of each model is evaluated using key metrics such as accuracy, precision, recall, and F1-score. Additionally, confusion matrices and graphical visualizations are used to analyze model behavior. The proposed Adaptive Distance Weighted KNN model demonstrates improved performance over traditional approaches.

The results indicate that adaptive techniques significantly enhance classification accuracy and reduce misclassification.

Chapter 1

Introduction

1.1 History

Machine Learning enables systems to learn from data and make predictions. K-Nearest Neighbors (KNN) is a simple and widely used classification algorithm.

1.2 Requirement Analysis

The project requires a dataset, Python programming environment, and libraries such as NumPy, Pandas, Matplotlib, and Scikit-learn.

1.3 Main Objective

To improve the performance of the KNN algorithm using adaptive distance-based weighting techniques.

1.4 Sub Objectives

- Implement baseline KNN
- Apply distance weighted KNN
- Develop adaptive KNN
- Compare model performance

1.5 PERT Chart Legend

Project phases include data preprocessing, model building, evaluation, and analysis.

Chapter 2

System Analysis

2.1 Existing System

Traditional KNN treats all neighbors equally, leading to poor performance in noisy datasets.

2.2 Motivations

To improve classification accuracy by assigning weights based on distance and dynamically selecting neighbors.

2.3 Proposed System

The system includes baseline KNN, weighted KNN, and adaptive KNN for improved performance.

2.4 Modules

- Data Preprocessing
- Feature Selection
- Model Training
- Evaluation
- Visualization

Chapter 3

Design

3.1 System Design

The system follows a pipeline: data preprocessing → feature selection → model training → evaluation.

3.2 Architecture

Input Data → Preprocessing → Model Training → Evaluation → Output

Chapter 4

KNN Classification System

4.1 Overview

This system uses KNN-based models for classification tasks.

4.2 Working

The model calculates distances between data points and predicts class based on nearest neighbors.

4.3 Technologies Used

- Python
- NumPy
- Pandas
- Scikit-learn
- Matplotlib

Chapter 5

Implementation

5.1 Methodology

Data is preprocessed, features are selected, and models are trained and evaluated.

5.2 Algorithms

- K-Nearest Neighbors (KNN)
- Distance Weighted KNN
- Adaptive Distance Weighted KNN

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Chapter 6

Output Screens

6.1 Feature Distributions

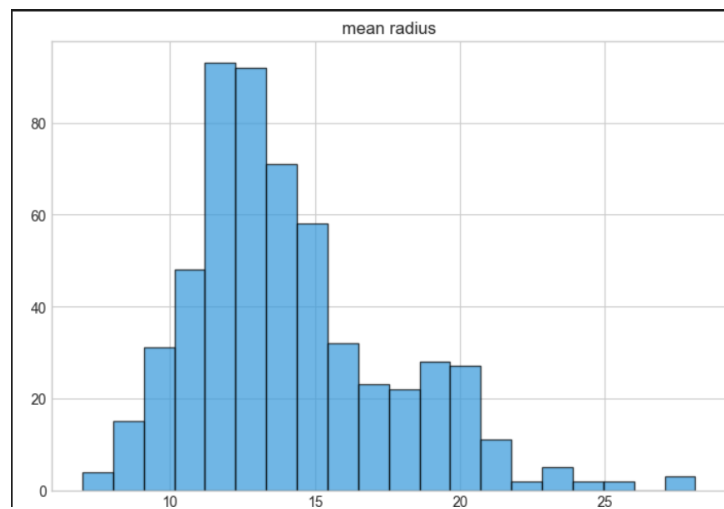


Figure 6.1: Distribution of Mean Radius

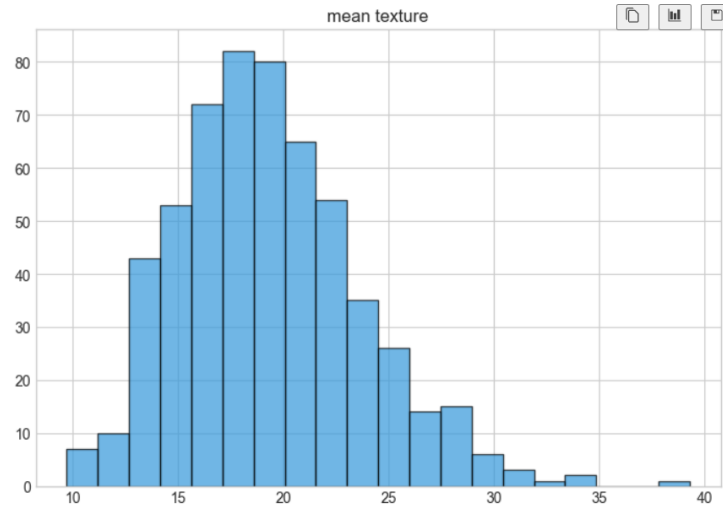


Figure 6.2: Distribution of Mean Texture

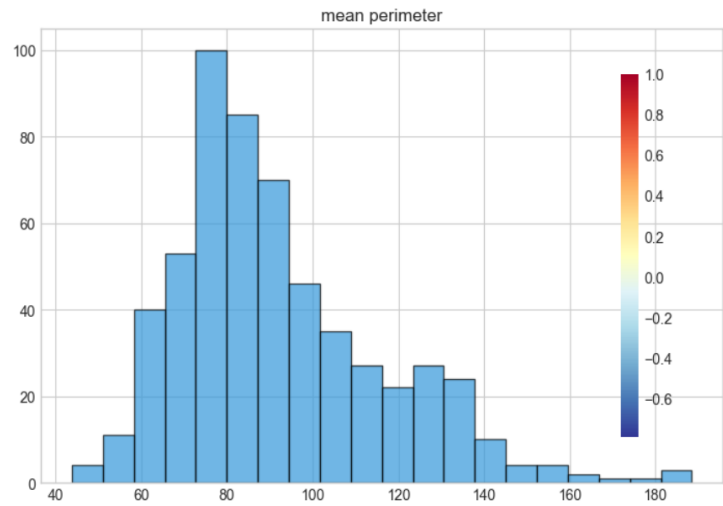


Figure 6.3: Distribution of Mean Perimeter

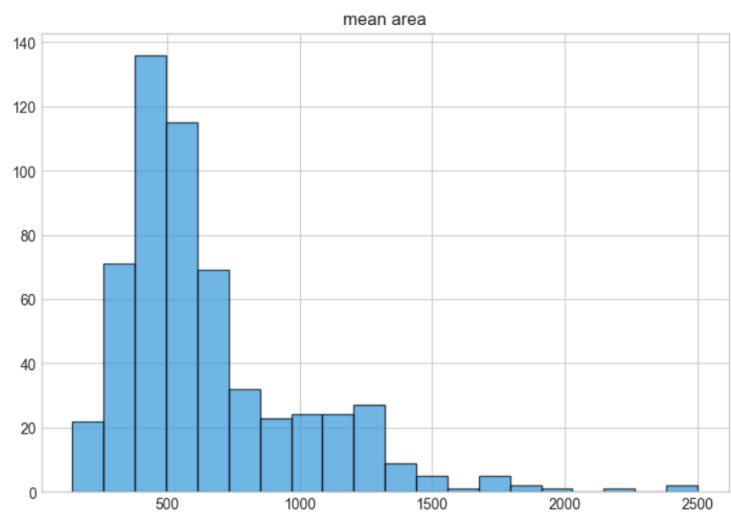


Figure 6.4: Distribution of Mean Area

6.2 Model Performance

```
=====
BASELINE KNN (K=5, Uniform Weights)
=====
```

✅ Accuracy: 0.9561

📊 Confusion Matrix:

```
[[39  3]
 [ 2 70]]
```

📄 Classification Report:

	precision	recall	f1-score	support
Malignant	0.95	0.93	0.94	42
Benign	0.96	0.97	0.97	72
accuracy			0.96	114
macro avg	0.96	0.95	0.95	114
weighted avg	0.96	0.96	0.96	114

Figure 6.5: Baseline KNN Performance

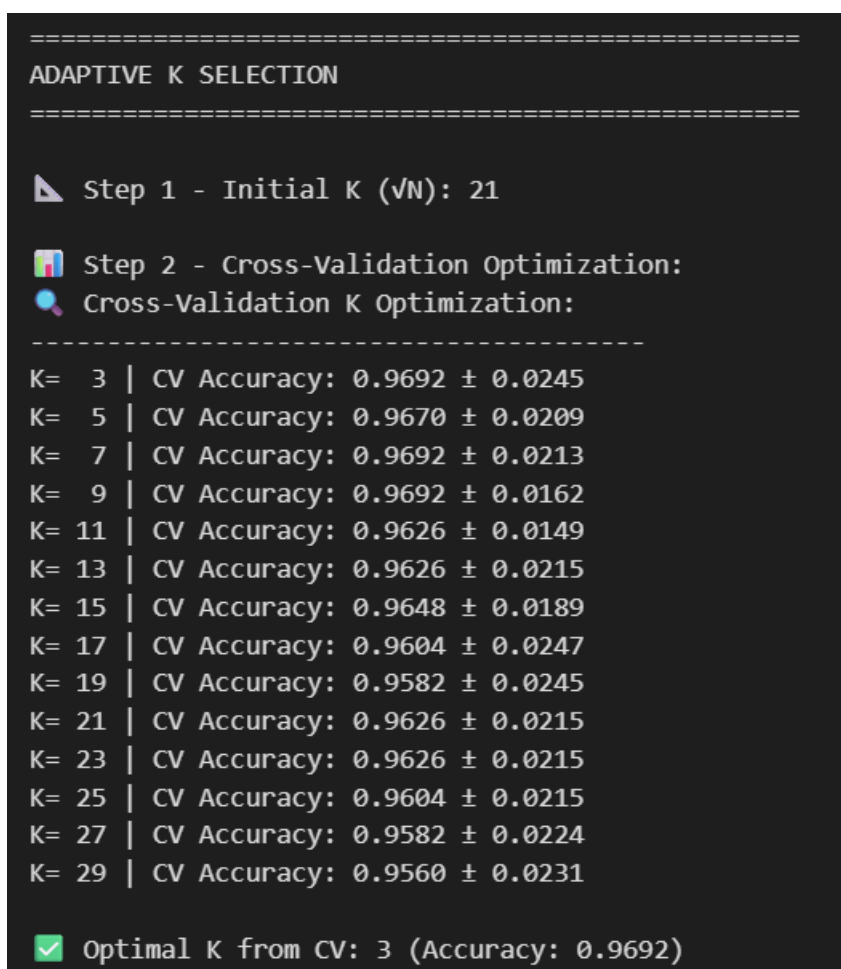


Figure 6.6: Adaptive K Selection using Cross Validation

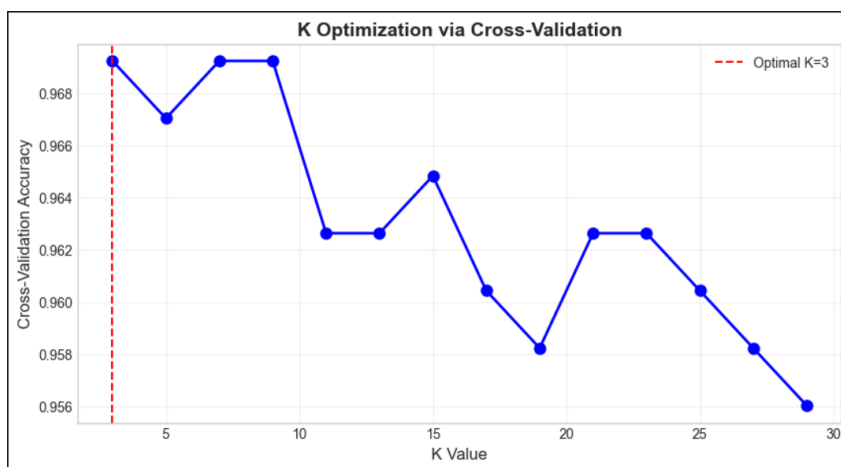


Figure 6.7: K Optimization Graph

6.3 Final Model Results

```
=====
FINAL MODEL: ADAPTIVE DISTANCE WEIGHTED KNN
=====

✅ Accuracy: 0.9825
📈 Improvement over baseline: 2.63%
📈 Improvement over weighted: 2.63%

📊 Confusion Matrix:
[[40  2]
 [ 0 72]]

📄 Classification Report:

```

	precision	recall	f1-score	support
Malignant	1.00	0.95	0.98	42
Benign	0.97	1.00	0.99	72
accuracy			0.98	114
macro avg	0.99	0.98	0.98	114
weighted avg	0.98	0.98	0.98	114

Figure 6.8: Final Adaptive Distance Weighted KNN Results

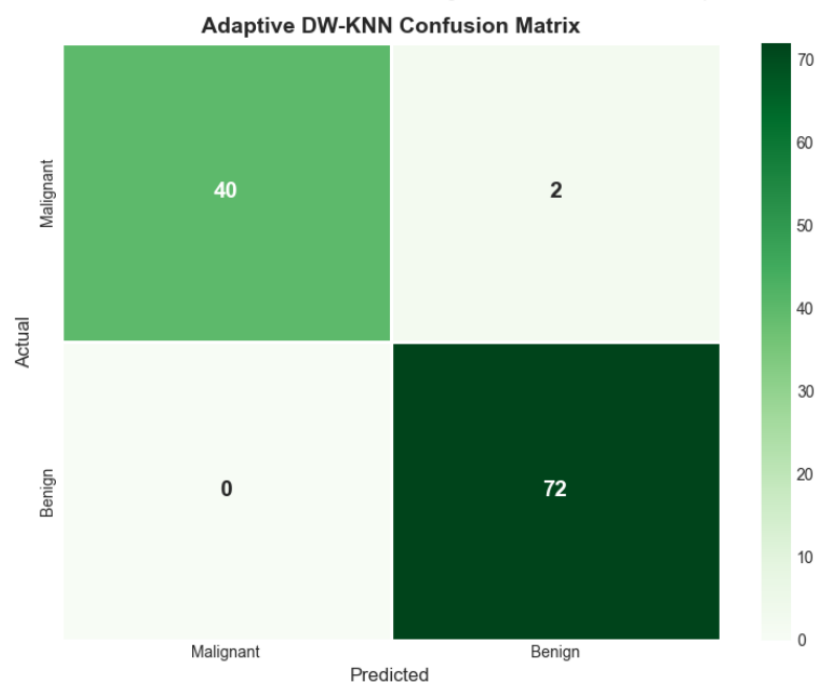


Figure 6.9: Confusion Matrix of Final Model

6.4 Performance Comparison

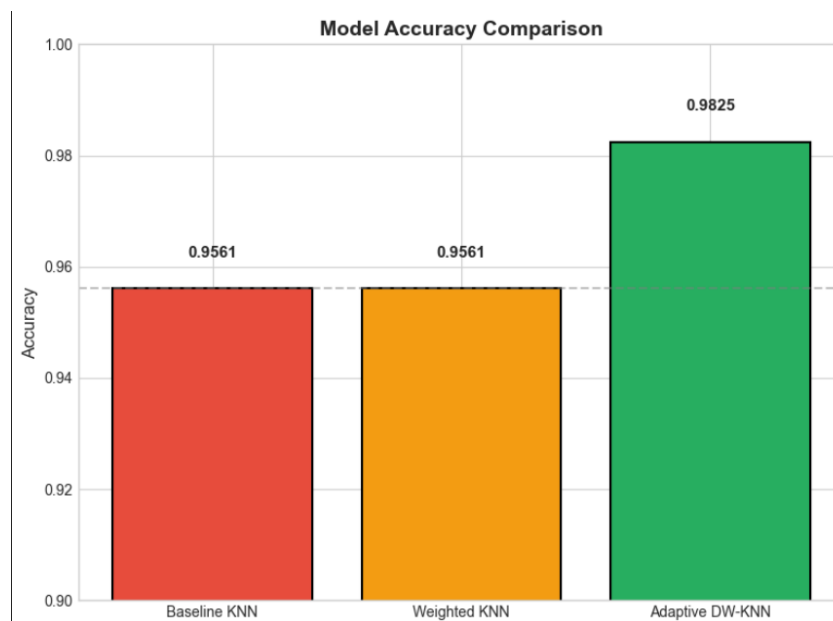


Figure 6.10: Accuracy Comparison of Models

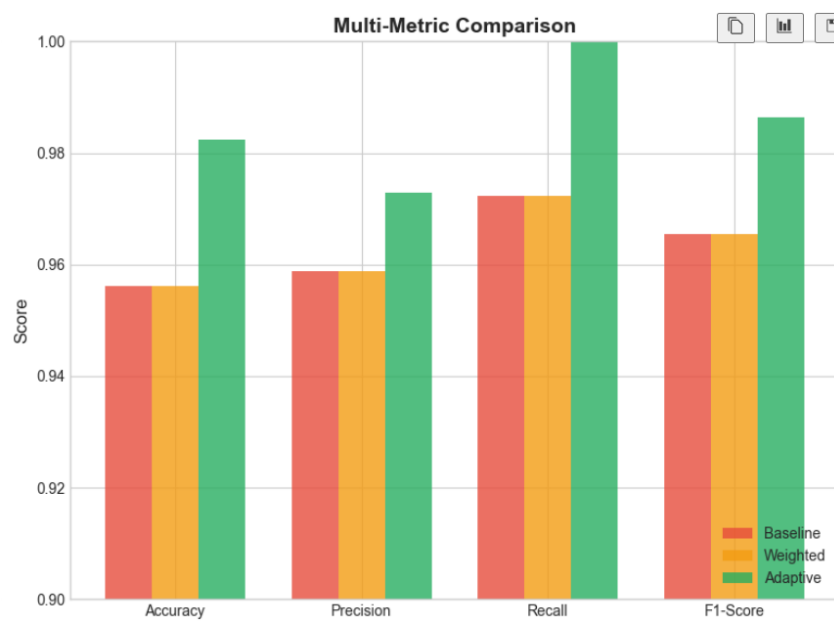


Figure 6.11: Multi-Metric Comparison (Precision, Recall, F1-Score)

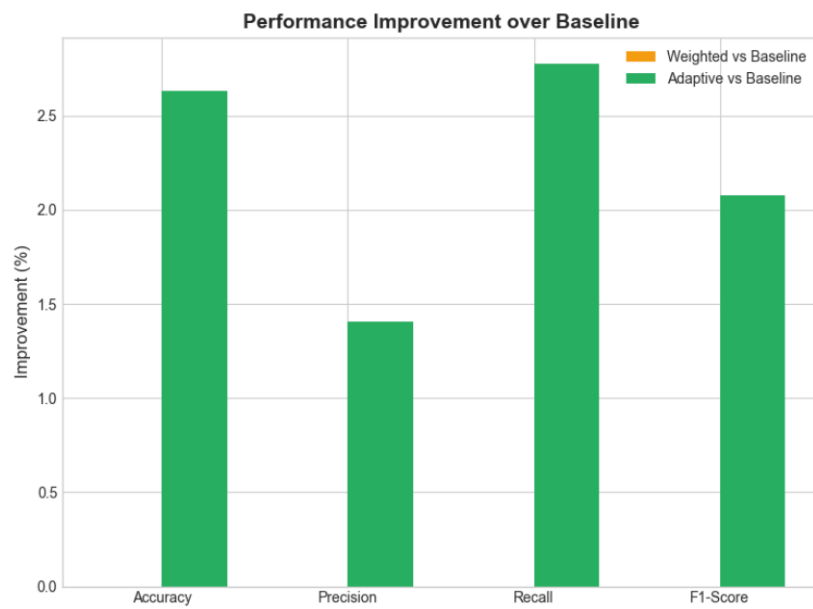


Figure 6.12: Performance Improvement over Baseline

6.5 Comparative Confusion Matrix Analysis

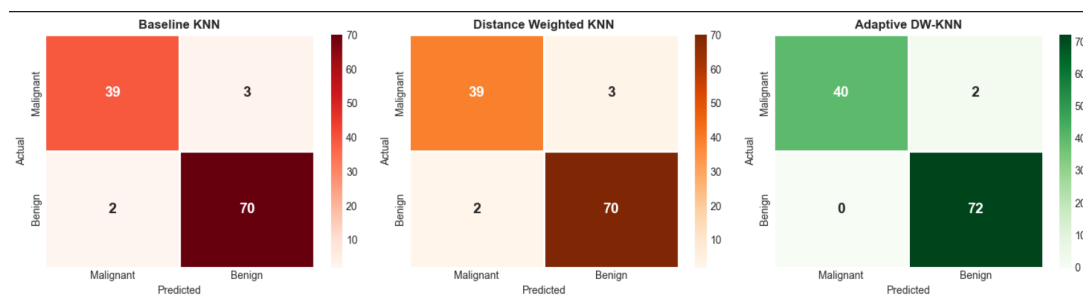


Figure 6.13: Comparison of Confusion Matrices for Baseline KNN, Distance Weighted KNN, and Adaptive Distance Weighted KNN

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Chapter 7

Limitations and Future Enhancements

Limitations:

- Sensitive to dataset size
- Computationally expensive

Future Enhancements:

- Use larger datasets
- Apply deep learning models
- Improve parameter tuning

Chapter 8

Conclusion

The project demonstrates that adaptive techniques improve KNN performance. The proposed model achieved better accuracy and reduced misclassification compared to traditional approaches.

Chapter 9

References

- DW-KNN: A Transparent Local Classifier Integrating Distance Consistency and Neighbor Reliability
- Adaptive Weighted Approach for High-Dimensional Statistical Learning and Inference
- Effective K-Nearest Neighbor Models for Data Classification Enhancement
- Double Weighted K-Nearest Neighbours for Binary Classification of High Dimensional Genomic Data
- Scikit-learn Documentation: <https://scikit-learn.org/>
- Python Documentation: <https://docs.python.org/>

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Plagiarism Report

This is to certify that the project report titled “**Adaptive Distance Weighted K-Nearest Neighbors for High-Dimensional Classification**” submitted by us is an original work and has been carried out under the guidance of our mentor.

We hereby declare that this report has been checked using standard plagiarism detection tools and the similarity index is within the permissible limits as prescribed by the university guidelines.

The similarity index of this report is below 15% (excluding references and citations), and the AI-generated content percentage is below 20%.

All sources of information have been properly acknowledged and cited in the bibliography section.

Ananya Chakravorty 500121827

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Mentor Signature

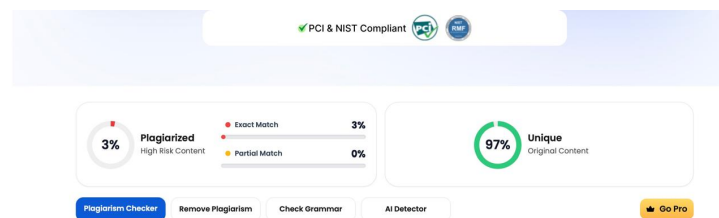


Figure 9.1: Plagiarism Report Summary

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Appendix A

Dataset Description

The dataset used in this project is a binary classification dataset used to distinguish between two classes (malignant and benign cases). It consists of multiple numerical features representing characteristics of the data samples.

Dataset Details:

- Number of samples: 569
- Number of features: 30
- Target variable: Binary (0 = benign, 1 = malignant)

Sample Data:

Feature1	Feature2	Feature3	...	Target
12.3	45.6	78.9	...	1
10.2	40.1	70.5	...	0

The dataset was preprocessed using normalization and feature selection techniques before training the models.

Appendix B

Code Snippets

This section provides key parts of the implementation used in the project.

K-Nearest Neighbors Model:

```
from sklearn.neighbors import KNeighborsClassifier

knn = KNeighborsClassifier(n_neighbors=5)
knn.fit(X_train, y_train)
y_pred = knn.predict(X_test)
```

Weighted KNN Logic:

```
weights = 1 / (distances + 1e-5)
```

Evaluation Metrics:

```
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score

accuracy = accuracy_score(y_test, y_pred)
```


Appendix C

Mathematical Formulation

The following formulas are used in the implementation:

Euclidean Distance:

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

Weighted KNN:

$$w_i = \frac{1}{d(x, x_i)}$$

Prediction Rule:

$$\hat{y} = \arg \max \sum w_i \cdot y_i$$

These formulas are used to compute distances and assign weights to neighbors for classification.

Appendix D

Additional Results

This section includes additional outputs and visualizations generated during the project.

Included Results:

- Confusion matrices of all models
- Accuracy comparison graphs
- Feature distribution plots
- Correlation heatmap